

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

Let's break down the problem step by step to find out how much Kylar needs to pay for 16 glasses .

1 . ** Determine the cost of the first and every second glass : **

- The cost of the first glass is \$ 5 .
- The cost of every second glass is 60 % of \$ 5 .
- To find the cost of every second glass :

$$0.60 \times 5 = 3 \text{ dollars}$$

So , the cost of every second glass is \$ 3 .

2 . ** Identify the pattern for the 16 glasses : **

- Kylar is buying 16 glasses .
- According to the pattern , every second glass costs \$ 3 , starting from the second glass .
- This means the glasses will be grouped as : First glass , Second glass , Third glass , Fourth glass , and so on , with the second , fourth , sixth , and so on , costing \$ 3 , while the first , third , fifth , and so on , costing \$ 5 .

3 . ** Count how many glasses cost \$ 5 and how many cost \$ 3 : **

- For 16 glasses , the pattern will be : 1 , 2 , 3 , 4 , 5 , 6 , 7 , 8 , 9 , 10 , 11 , 12 , 13 , 14 , 15 , 16 .
- The glasses that cost \$ 5 are the 1 st , 3 rd , 5 th , 7 th , 9 th , 11 th , 13 th , and 15 th glasses . That’s 8 glasses .
- The glasses that cost \$ 3 are the 2 nd , 4 th , 6 th , 8 th , 10 th , 12 th , 14 th , and 16 th glasses . That’s also 8 glasses .

4 . ** Calculate the total cost : **

- Cost for the 8 glasses at \$ 5 each :
- $$8 \times 5 = 40 \text{ dollars}$$
- Cost for the 8 glasses at \$ 3 each :
- $$8 \times 3 = 24 \text{ dollars}$$
- Adding these two amounts together :
- $$40 + 24 = 64 \text{ dollars}$$

<think>

- We first identified the price for the first and every second glass .

- We then determined the pattern and counted the number of glasses at each price .

- Finally , we calculated the total cost by adding the total costs of the glasses priced \$ 5 and the glasses priced \$ 3 .

</think>

<answer>

64 dollars

</answer>